Klimate Kundli

Clarifications on the Proposal

Klimate Kundli is based on the Indian astrological birth chart or kundli.
A kundli is a snapshot of 12 planetary positions in a moment in time (i.e. one's birth) which is then used to make predictions about one's life experiences and future. With climate being a relatively individual experience, we saw the structure of a kundli as a helpful way of presenting a snapshot and possible futures of an individual's climate experiences.

<u>How the "Klimate Kundli" would be visualised</u>

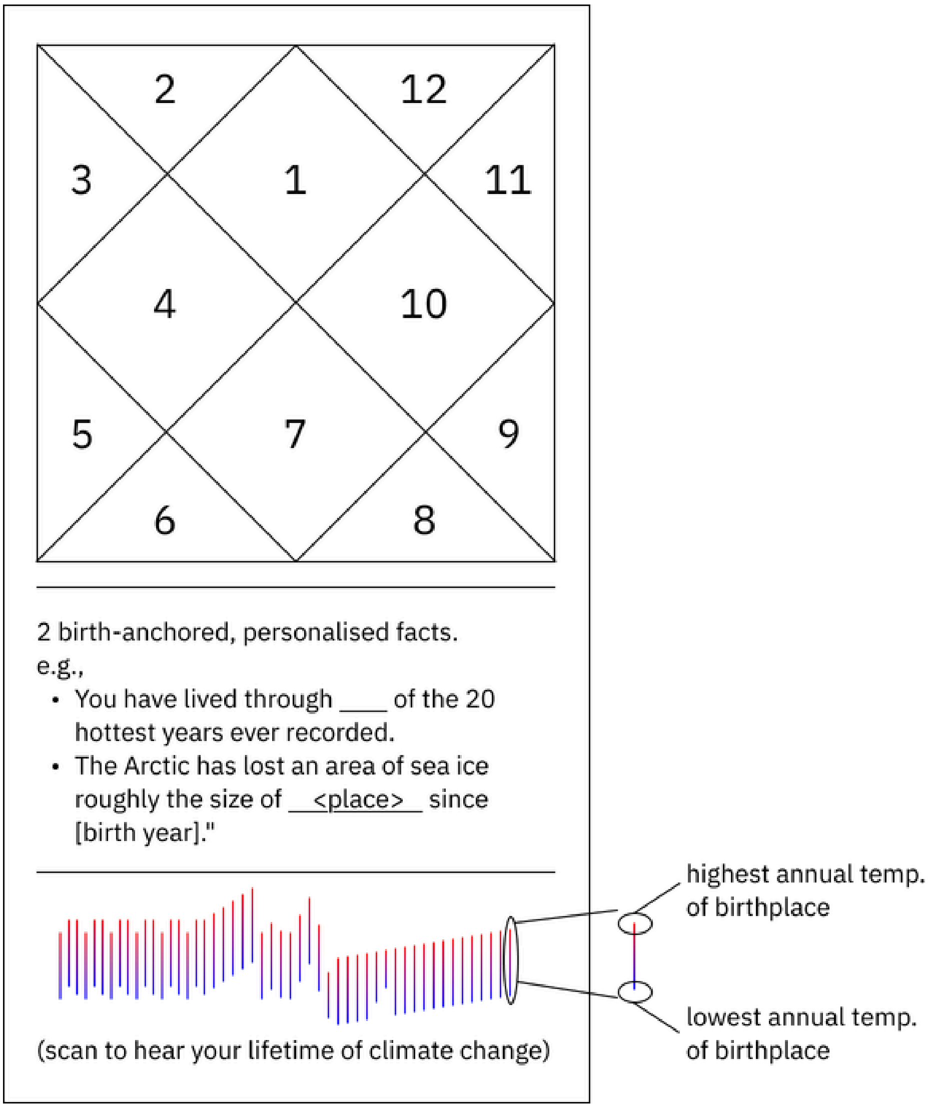
The following page contains a mock-up of the Kundli Card that will be printed.
We aim to have the following data showcased, mapped to each corresponding number:

1. Birth date + city of birth
2. Highest temp experienced in birth year
3. Lowest temp experienced in birth year
4. Highest temp of your birth place on your latest birthday
5. Lowest temp of your latest birthday
6. Range of variation in summer temps across cities inhabited
7. Range of variation in winter temps across cities inhabited
8. Change in national carbon emissions lifetime
9. Change in rainfall in birth city lifetime
10. Sea-level rise in the world lifetime
11. Comparing PPM/CO2 levels
12. Projected temp in 2050

This list is *not* exhaustive as we are continually iterating and considering use cases to see which data points have existing public data sets we can refer to and would be interesting for folx to experience.
The order of presentation intends to take the user from personal data points to larger ecological climate histories and eventually future predictions. There are also two interesting facts about the larger climate history that will be printed under the kundli.

<u>How the comparison between remembered climate and present-day conditions would appear in the printed artefact</u>

To generate the Kundli, a user takes a quiz, which asks them to input their name, place of birth, birth date and also map the cities they have lived in for more than 6 months and the durations. The comparisons being drawn between climate experiences is anchored in places inhabited by the user.
One consideration we are thinking with since we received this question from you is whether, as part of this quiz, we can ask users to reflect on the hottest and coldest temperatures they remember experiencing. For example, Nithya recalls experiencing 0°C in Delhi once when she was 5-6 years old. Does this align with historical data? What does it say about temperature as being felt and not just calculated and the ramifications on how we calculate warming?



An added element, if we figure out a way to make it work.

P.S: The “barcode” at the bottom of your kundli is a musical score. Each vertical line is one year of your life. Lines extending upward mark the highest recorded temperature in your birth city that year. Lines extending downward mark the lowest. Their color follows the actual temperature — blue for cold years, through green and yellow for moderate ones, into orange and red as things get hotter. A city with mild winters and scorching summers might show green below and deep red above in the same year.
A program maps these temperatures to pitch — hotter years sound higher, colder years lower. The two notes for each year play simultaneously as a chord. When the range between them is narrow, the chord sounds stable. When extremes pull apart, it gets tense, dissonant.
Scan the code. You'll hear your birth city's climate, year by year, from the year you were born until now. Twenty to sixty seconds of audio, depending on your age. The early years tend to sit lower, steadier. The recent ones drift up.

How the visitor journey work across the station and printed artefact

The station is a table with an iPad, a TV, and a printer. On the iPad, visitors enter their birth date, birth city, current city and how long they've lived there, and any other cities they've passed through, with approximate years. This could take about 90 seconds.
While the card generates (2–4 seconds, live query), the TV shows a temperature anomaly timeline for their birth city, pulled from the same data. The TV gives the city's arc; the card gives the visitor's position within it.
The card has three layers. The kundli grid at the top, with twelve climate data points mapped to twelve houses. Two birth-anchored facts in the middle, printed as plain text. And a colored barcode at the bottom. Visitors scan it on their phones and listen with the card still in hand.
The card is designed to be compared. People hold them next to each other, point at different cells, ask where the other person grew up. The barcode lengths alone (visibly shorter for younger visitors, wider for older ones) start that comparison without words. Asking visitors how old they will be that year tends to shift the exchange from data to something more personal. The audio adds a second entry point: two people scanning their barcodes next to each other hear their cities diverge.

What data sources you plan to use, and how feasible the proposed data access and processing will be.

1. Primary API: Open-Meteo Historical Weather API

Free, no authentication, global coverage back to 1940. Uses ERA5 reanalysis under the hood. Accepts latitude/longitude and returns daily temperature (min, max, mean) and precipitation as JSON. Handles Indian cities and foreign cities identically. Response times are under 2 seconds for a 50-year range.

GET https://archive-api.open-meteo.com/v1/archive
?latitude=28.61&longitude=77.20
&start_date=1985-01-01&end_date=2025-12-31
&daily=temperature_2m_max,temperature_2m_min,precipitation_sum

This single endpoint covers most of the kundli's data needs: birth day temps, latest birthday temps, summer/winter ranges, rainfall trends, and the full annual high/low series for the barcode sonification.

2. Geocoding: Open-Meteo Geocoding API
Also free, no auth. Converts a city name to lat/lon. Handles Indian cities, including smaller ones, and international cities.

GET https://geocoding-api.open-meteo.com/v1/search?name=Bhopal&count=1

3. For carbon emissions: Our World in Data / Global Carbon Project
Country-level annual CO2 emissions as a static CSV download (~200KB), updated yearly. Pre-bundle this. No API call needed — a lookup by country code and year is instant.

4. For sea-level rise: NASA Sea Level dataset
A single global time series. Also static, updated quarterly. Pre-bundle it.

5. For 2050 projections: CMIP6 via Copernicus CDS API
Requires a free account and an API key. Returns gridded projections by scenario (SSP2-4.5 is the moderate path, SSP5-8.5 is worst-case). Query by lat/lon. This one is slower (5–15 seconds) and worth pre-caching.

Script logic

Visitor types in city of birth and birth date. The script geocodes the city, then fires two parallel requests to Open-Meteo: one for the full daily temperature/precipitation history (birth year to 2025), one for the CMIP6 2050 projection if it's not already cached. While those return, it looks up carbon emissions and sea level from bundled CSVs using the country code and birth year.

From the daily data, the script computes: max and min temp on their exact birth date, max and min temp on the same calendar date in 2025, summer and winter temperature ranges per decade (to show how the spread has shifted), annual rainfall trend, and the year-by-year high/low series for the barcode.
All twelve kundli cells and the sonification data are derived from these computations. Generation takes 2–4 seconds from input to output.

Live query vs. pre-built table

Pre-building is not feasible. Visitors might be born in Coimbatore, Karachi, Nairobi, or Zurich. You cannot anticipate every city. Open-Meteo is fast enough for live use and has no rate limits for reasonable traffic.

What you should pre-cache: the static datasets (carbon, sea level), the CMIP6 projections for commonly expected cities (top 50–100 Indian cities plus 20–30 major international ones), and a local results cache that stores computed kundli data by city+birthdate. If the same city appears twice in a day, the second lookup is instant. Over the course of an exhibition, the cache fills itself with the long tail.

If the venue's internet drops, the cache covers most visitors. A fallback message for genuinely uncached cities ("your kundli is being prepared: collect it in 5 minutes") buys time.